

Space Stone

Technical Sheet — Participant Guide

Data Space Club — ENSSEA
Geographic Market Listings Dataset

Quick Summary

Silhouette Score: 60% | **Application Quality:** 30% | **Presentation:** 10%

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1 The Problem

You are given a dataset of 15,000 records representing real-world market listings. Each record has geographic coordinates and various market-related features (prices, availability, categories, etc.).

Your mission: Discover the natural geographic clusters hidden in this data — without any labels to guide you.

Why this matters: In real business scenarios, companies often have customer/location data without predefined segments. Identifying natural clusters helps with:

- Targeted marketing campaigns.
- Optimal store placement.
- Pricing strategy per region.
- Resource allocation.

The challenge: You don't know:

- How many clusters exist.
- What features define the clusters.
- Which features are noise.
- Whether the coordinates are Euclidean or geographic.

2 The Dataset

2.1 Overview

You are provided with a single CSV file containing 15,000 rows and 15 columns. The data is real, derived from actual market listings.

Important

The column names have been anonymized and renamed. You must figure out what each column represents through exploratory data analysis (EDA). Some columns may be noise (random data with no real meaning) — identify and exclude them.

2.2 Columns Provided

2.3 Data Characteristics

- **Missing values:** 10–15% of cells contain NaN (varies by column).
- **Outliers:** Some numeric columns contain extreme values.

Column	Type	Description
`id`	int	Unique record identifier
`coord_x`	float	Coordinate X (interpretation unknown)
`coord_y`	float	Coordinate Y (interpretation unknown)
`metric_A`	float	Numeric metric (may have outliers)
`metric_B`	float	Numeric metric
`metric_C`	float	Derived metric
`category_R`	str	Categorical (3 types)
`count_N`	int	Count metric
`min_nights`	int	Minimum value
`reviews_freq`	float	Frequency metric
`host_count`	int	Count metric
`ratio_AB`	float	Derived ratio
`distance_unknown`	float	Distance metric
`noise_0 - noise_4`	float	5 features of unknown nature

- **Noise features:** At least 5 columns are pure noise (random).
- **Mixed types:** Numeric, categorical, and derived features.
- **Unknown K:** The number of clusters is not provided.

3 Files Provided

File	Description
`space.csv`	The dataset (15,000 rows × 15 columns)
`sample_submission.csv`	Submission format template
`technical_sheet.pdf`	This guide

4 The Evaluation — Haversine Silhouette Score

4.1 Why Haversine?

The Earth is spherical, not flat. Standard Euclidean distance assumes a flat plane, which gives incorrect results for geographic coordinates. The Haversine formula computes the great-circle distance between two points on a sphere, giving the true geographic distance.

4.2 The Haversine Formula

Info

Haversine Distance

$$\text{Haversine}(\phi_1, \lambda_1, \phi_2, \lambda_2) = 2R \arcsin \sqrt{\sin^2\left(\frac{\phi_2 - \phi_1}{2}\right) + \cos(\phi_1)\cos(\phi_2)\sin^2\left(\frac{\lambda_2 - \lambda_1}{2}\right)}$$

where:

- ϕ = latitude (in radians)
- λ = longitude (in radians)
- R = Earth's radius (≈ 6371 km)

4.3 The Silhouette Score

For each point i:

- a_i = average Haversine distance to all other points in the same cluster (compactness).
- b_i = average Haversine distance to all points in the nearest other cluster (separation).

Info

Silhouette Score Formula

$$S_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

$$\text{Final Score} = \frac{1}{n}\sum_{i=1}^{n} S_i$$

4.4 Score Interpretation

Score Range	Quality	Meaning
0.7 – 1.0	Excellent	Strong, well-separated clusters
0.5 – 0.7	Good	Reasonable structure found
0.25 – 0.5	Fair	Weak structure, could be improved
0.0 – 0.25	Poor	Clusters overlap significantly
< 0.0	Bad	Worse than random assignment

4.5 The Black Box Leaderboard

Black Box Scoring

The leaderboard is completely hidden during the event. No team can see other teams' scores. The final ranking will be revealed only at the closing ceremony.

This means:

- You cannot overfit to the leaderboard.
- You cannot copy strategies from higher-ranked teams.
- Your score is a surprise until the end.

5 Scoring Distribution

Your final grade is composed of three parts:

Component	What's Evaluated	Weight
Haversine Silhouette	Quality of your cluster assignments	60%
Application Quality	Code, EDA, GitHub repo, pipeline design	30%
Presentation	Live explanation of your clusters	10%

6 Submission Format

You must submit a CSV file named `submission.csv` containing your cluster assignments.

6.1 CSV Format

```
id,cluster 1,0 2,2 3,1 4,0 5,3
```

6.2 Rules

- `id` must match the dataset exactly.
- `cluster` must be an integer ≥ 0.
- You must assign a cluster to every record.
- The number of clusters is your choice (recommended: 3–7).
- Do not submit your code in this file — only cluster assignments.

7 Conditions and Rules

7.1 Technology Requirements

Allowed Technologies

- **Clustering algorithms:** K-Means, DBSCAN, HDBSCAN, Hierarchical, Spectral, Mean Shift, Gaussian Mixture, OPTICS, Affinity Propagation.
- **Dimensionality reduction:** PCA, t-SNE, UMAP.
- **Libraries:** scikit-learn, scipy, hdbscan, umap-learn.

- **Visualization:** matplotlib, seaborn, folium, plotly.
- **Languages:** Python (recommended).

NOT Allowed

- No external datasets (only the provided `space.csv`).
- No pre-trained models.
- No paid APIs.
- No Deep Learning for clustering (autoencoders, deep clustering).

7.2 Notebooks and Code Files

You must submit:

- Jupyter notebooks (`.ipynb`) showing your analysis.
- Python scripts (`.py`) for the final pipeline.
- Any visualization files (PNG, HTML).

Repository Naming

Name your repository: `space-stone-team-X` (replace X with your team number).

8 Suggested Workflow

Phase	Task	Time
1	Load data, understand columns	1 hour
2	EDA: distributions, correlations, missing values	1.5 hours
3	Identify noise features (drop them)	30 min
4	Determine if coords are geographic or Euclidean	30 min
5	Handle missing values + outliers	1 hour
6	Feature engineering + scaling	1 hour
7	Try multiple clustering algorithms	2 hours
8	Find optimal K (elbow, silhouette)	1 hour
9	Visualize clusters on a map	1 hour
10	Generate submission + GitHub repo	1 hour
Total		**10 hours**

9 Tips for Success

- **Investigate the coordinates:** If `coord_x` and `coord_y` are geographic (lat/long), use Haversine distance. If Euclidean, use standard distance.

- **Drop noise features:** Use correlation analysis to identify the 5 random columns.
- **Handle missing values:** Don't just drop rows — impute intelligently.
- **Outliers:** Some are real (luxury listings), some are errors. Decide carefully.
- **Scaling matters:** Standardize numeric features before clustering.
- **Try multiple algorithms:** K-Means assumes spherical clusters; DBSCAN finds arbitrary shapes.
- **Visualize:** Plot the clusters on a geographic map to see if they make sense.
- **Choose K wisely:** Use elbow method + silhouette analysis + domain intuition.

10 Submission Checklist

Before submitting, verify:

☐ `submission.csv` follows the exact format.

☐ `id` matches the dataset.

☐ Every record has a cluster assignment.

☐ Cluster values are integers ≥ 0.

☐ You chose a reasonable K (3–7).

☐ Notebooks are clean and reproducible.

☐ You identified and dropped noise features.

11 Evaluation Summary

Component	Description	Weight
Haversine Silhouette Score	(Black Box)	60%
Application Quality	(Code + EDA)	30%
Presentation	(Live Cluster Explanation)	10%

Good luck! Discover the hidden geographic structure of the market.

Space Stone — PULSAR Hackathon, Data Space Club, ENSSEA